Estimating the Forage Neutral Detergent Fiber Content of Alpine Grassland in the Tibetan Plateau Using Hyperspectral Data and Machine Learning Algorithms

Jie Liu, Qisheng Feng, Tiangang Liang, Jianpeng Yin, Jinlong Gao, Jing Ge, Mengjing Hou, Caixia Wu, and Wenlong Li

***Abstract*— The neutral detergent fiber (NDF) content is a key factor in the forage palatability and livestock digestibility of alpine grassland. Traditional methods of measuring NDF are time-consuming and laborious, while multispectral remote sensing often has difficulty capturing subtle changes due to its broad spectral channels. In this study, we evaluate NDF using hyperspectral data and environmental factors (EFs) during 2016–2019. The artificial neural network (ANN), support vector machine (SVM), and random forest (RF) are used to construct models. The results show that: 1) compared with the correlation analysis, the least absolute shrinkage and selection operator (LASSO) regression can more effectively select the important variables for estimating NDF and achieve significant dimension reduction; 2) estimation accuracy of the models (coefficient of determination (R^2) between 0.44 and 0.51, relative root mean square error (RMSE) between 4.31% and 4.61%) based on first derivative (FD) spectra is relatively better than the models based on original spectra (OR), log transformation (Log), and continuum removal (CR) spectra (R^2 between 0.35 and 0.46, RMSE between 4.39% and 4.92%). The RF model based on FD-Log-vegetation indices (VIs)-EFs is the best model (R^2 = 0.62, RMSE = 3.98%); and 3) compared with the EFs, hyperspectral feature plays a pivotal role in estimating NDF, with a total contribution of 81.66%, and red-edge and shortwave infrared (SWIR) regions are significant for estimating NDF. In general, the combination of hyperspectral data with the EFs significantly improves the accuracy of estimating NDF. This study provides important guidance for future efforts on estimating NDF of natural grassland using remote sensing and machine learning algorithms.**

***Index Terms*— Alpine grassland, hyperspectral remote sensing, machine learning, neutral detergent fiber (NDF).**

I. Introduction

NATURAL grassland is the food supply base for grazing livestock and wild animals, and it provides a large amount of food, goods, and services for humans. To meet the growing demand, livestock and grassland management must be further improved to increase grassland productivity as well as commodity yield and quality [1], [2]. The palatability of forage and the digestibility of livestock seriously affect the feed intake of livestock, thereby affecting the level of livestock production [3], [4]. Neutral detergent fiber (NDF) in forage plays a crucial role in maintaining the normal digestion and metabolism of ruminant livestock [5], and it contains cellulose, hemicellulose, lignin, and silicate, among which cellulose and lignin are important for ecological processes (e.g., photosynthesis, transpiration, respiration, and decomposition) in plants and are highly correlated with the total carbon content of the leaves [6], [7]. Therefore, accurate and timely monitoring of changes in the content and spatial distribution characteristics of forage NDF is vital for the production and management of the grassland livestock industry as well as for the evaluation of the health of grassland ecosystems.

The hyperspectral remote sensing technique is an important approach for exploring high-accuracy inversion models for forage NDF content. Traditional monitoring and quantitative studies of the NDF mainly rely on chemical analysis and near-infrared (NIR) spectroscopy, both of which require destructive sampling. In particular, the wet chemical method is costly, time-consuming, and laborious [8]–[10]. In recent years, the non-destructive ground-based hyperspectral remote sensing technique has made good progress in the inversion of forage NDF with a prospect of widespread application [11]–[15]. Hyperspectral instruments have many consecutive narrow spectral channels, which can detect richer fine-spectral characteristics [16], [17]. Compared with the reflectance of the original spectra (OR) channels, the relevant vegetation

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indices (VIs), such as the simple ratio index (SR), normalized difference spectral index (NDSI), normalized difference lignin index (NDLI), and normalized difference vegetation index (NDVI), calculated on the basis of narrow spectral channels or transformed broad waveband reflectance, not only can improve the estimation accuracy of NDF but can also realize wide-range applications of the inversion models using the existing satellite remote sensing data. For example, Starks *et al.* [14] used a portable FieldSpec FR spectroradiometer (Analytical Spectral Devices Inc. (ASD), Boulder, CO, USA) to perform canopy reflectance spectral measurements and analyses of bermudagrass pastures in the Southern Great Plains of the USA. The OR spectral reflectance was transformed into six broad wavebands that corresponded to the Landsat Thematic Mapper (TM) bands. The correlations between single-band reflectance and NDF content and the correlations between VIs ($R_{\mathrm{NIR}}/R_{\mathrm{red}}$, $R_{\mathrm{blue}}/R_{\mathrm{red}}$, $R_{\mathrm{SWIR1}}/R_{\mathrm{red-edge}}$, $R_{\mathrm{NIR}}/R_{\mathrm{SWIR1}}$ and NDVI) with NDF were analyzed. The results showed that the correlations between most of the two-band reflectance ratios and NDF content were higher than those between the single band and NDF content, specifically, the correlation coefficient (r) between $R_{\mathrm{blue}}/R_{\mathrm{red}}$ and NDF content was the highest (-0.48). Zhao *et al.* [18] conducted canopy reflectance spectral measurements in the growing seasons of 2002–2003 of five fixed pastures [four monoculture-pastures with warm-season perennial bermudagrass and one mixed pasture (including bermudagrass and other warm-season grasses)] in the central Oklahoma of the USA. The reflectance data were averaged across 10 nm wavebands to obtain 120 narrow wavebands, and the NDF content was estimated by using the methods of multiple regression with maximum R^2 improvement (MAXR) and partial least square regression (PLSR). The results showed that compared with the PLSR method with 120 bands, the MAXR regression that contained 5–10 bands produced a similar coefficient of determination (R^2) (0.52 versus 0.58) and relative root mean square error (RMSE) (3.1% versus 2.92%) for prediction of the forage NDF. In addition, the R^2 of the MAXR multiple regression model for predicting NDF was 0.23 higher than that between the two-band reflectance ratio and the NDF content. Albayrak *et al.* [19] studied the spectral feature bands of the Vicia species in the Mediterranean region of Turkey using a portable spectrometer (ASD Inc.; Boulder, CO, USA) with a spectral band range of 325–1550 nm. $R_{\mathrm{NIR}}/R_{\mathrm{red}}$ and NDVI were calculated based on four broad wavebands [NIR (760–900 nm), red (630–690 nm), green (520–600 nm), and blue (450–520 nm)] transformed from the OR spectra, and the NDF contents of three different vetch species were found to be closely correlated with the VIs. $R_{\mathrm{NIR}}/R_{\mathrm{red}}$ and NDVI explained 61%–65% and 67%–72%, respectively, of the variation in the NDF contents of the vetch species. In addition to OR spectra and VIs, spectral transformation techniques that include first derivative (FD), continuum removal (CR), and log transformation (Log) not only can eliminate the effects of the soil background, solar angle, and atmosphere to enhance the forage nutrients absorption features but can also further optimize VIs [20]–[23]. For example, Albayrak [24] collected the spectral reflectance of sainfoin pasture in the Mediterranean region of Turkey during 2007–2008 using a portable spectroradiometer with spectral bands of 325–1150 nm. They found that the reflectance of the four most important wavebands of the FD (440, 530, 630, and 760 nm) was highly relevant with the NDF of sainfoin pasture, and the R^2 of their stepwise regression model reached 0.86. Guo *et al.* [11] studied the spectral characteristics of natural grassland in Grasslands National Park of Canada and the surrounding pastures. The park and the surrounding area consisted of needle-and-thread, blue grama, western wheatgrass, spikemoss, and junegrass. The results indicated that the transformation techniques of FD and CR greatly improved the relationships between the nutrient contents and the spectral reflectance, and the NDF content was not significantly correlated with the OR spectra, but the correlations of the FD and CR spectral reflectance with the NDF content were both as high as 0.5. Zeng and Chen [25] investigated the mixed grassland (dominated by intermediate wheatgrass, pubescent wheatgrass, and alfalfa) near the Montana State University Agricultural Research Center. The OR spectra were pre-processed by the FD transformation, and then used to build the inversion models of NDF at three growth stages (boot stage, peak growth, and dormancy) as well as the whole period using the methods of PLSR and stepwise multiple linear regression (SMLR). The model based on the SMLR method in the whole period could explain 78% of the variation in the forage NDF.

Of the methods for constructing inversion models, SMLR and PLSR are the most commonly used to monitor the contents of the fiber components and other nutrients in forage using hyperspectral remote sensing techniques [26]–[30]. Compared to traditional regression algorithms, non-parametric machine learning models, such as artificial neural network (ANN), support vector machine (SVM), and random forest (RF) algorithms have the advantages of avoiding multicollinearity, simple tuning, parallel processing, and fast computation speed. They can also process both discrete and continuous data to enhance model generalisability and noise-immunity capabilities [31]–[33]. Using the WorldView-2 satellite images and RF method, Ramoelo *et al.* [34] monitored the leaf nitrogen content of rangeland in the north-eastern part of South Africa and found that the RF model based on VIs could explain 89% of the variation in the leaf nitrogen. Using the natural grassland in the Fort Nottingham Nature Reserve, KwaZulu-Natal, South Africa, as research objects, Singh *et al.* [15] divided the samples into two parts: the validation data and training data, in the ratio of 7:3. The important variables were selected from OR spectra using the variable importance calculated by the RF, and, the NDF was estimated based on the known absorption bands and the important variables. The results for the RF model displayed a relatively good fit in terms of predicted and measured values ($R^2 = 0.54$). Singh *et al.* [35] collected samples of forage and ancillary data (leaf area index, chlorophyll, species types, and species count) from the African savanna (the main dominant species were Festuca costata, Tristachya leucothrix, and Themeda triandra) in South Africa, and estimated the NDF using a machine learning algorithm based on RapidEye-5 imagery (with the band range of 440–850 nm). The results indicated that the RF successfully predicted the NDF content, with an R^2 value

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of 0.74. Gao *et al.* [36] constructed models of phosphorus content using hyperspectral data and multiple ecological-environmental factors (i.e., meteorology, vegetation, soil, and topography) in the eastern Tibetan Plateau. They found that the SVM model based on the FD spectra and important factors could well predict the phosphorus content, and the R^2 of the validation set was 0.67. In summary, the use of hyperspectral data and machine learning algorithms in combination with environmental factors (EFs) is the fundamental way to construct high-accuracy inversion models of vegetation fiber components contents. However, existing studies still have the following problems and challenges.

1) Although machine-learning algorithms are computationally fast, they can make the model more complicated and computationally intensive in the presence of many variables and even reduce the robustness and prediction accuracy of the model to some extent. Therefore, the selection of variables is an important prerequisite and foundation for establishing high-accuracy models to estimate the contents of the fiber components, which not only plays a crucial part in reducing the time needed for model development and application, but also improves the interpretability and stability of the model.
2) The commonly used machine learning algorithms, including RF, ANN, and SVM, work on different principles. In terms of the inversion models for the NDF content, it is still necessary to investigate which method is best under the influence of factors such as different regional ecological environments and the biochemical characteristics of the research objects.
3) Although some studies have been conducted to monitor NDF, acid detergent fiber (ADF), and cellulose contents using hyperspectral remote sensing data, few studies have combined spectral absorption features with VIs and EFs to establish models of the NDF in the alpine grassland.

Given the findings described above, this study combines the spectral absorption feature parameters of hyperspectral remote sensing data (including transformed spectral data) with EFs and uses machine learning algorithms to explore methods for remote-sensing monitoring of alpine grassland forage NDF content in the east of the Tibetan Plateau. The main objectives are: 1) to analyze the spectral bands, VIs, and EFs related to forage NDF content and to select the most sensitive variables to NDF and 2) to evaluate the accuracy of inversion models constructed using three non-parametric models (RF, ANN, and SVM) in estimating forage NDF content and to determine the best model for estimating the forage NDF content. This study is expected to provide technical support for the accurate evaluation of the vegetation growth status of alpine grassland.

II. Data and Methods

A. Study Area

The study area (33.10 °N–35.60 °N, 100.76 °E–104.76 °E), which is located in the northeast margin of Tibetan Plateau (Fig. 1), has a typical plateau continental climate with an average elevation of 3000 m, and it is cold and wet all year,

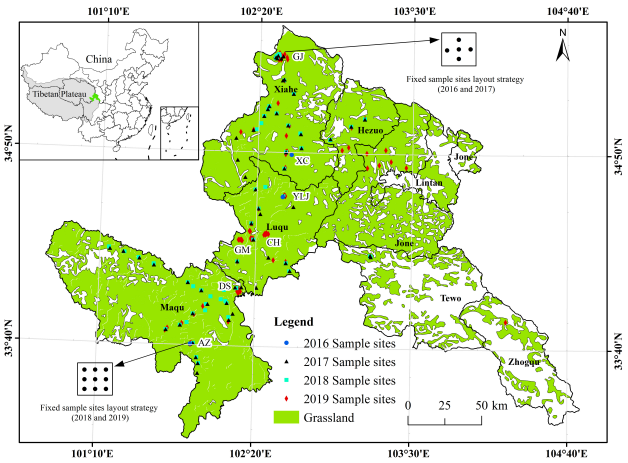


Fig. 1. Distribution of the permanent and random monitoring areas throughout the study area during 2016–2019.

with an average annual temperature of only 1 °C to 13 °C. The precipitation, with an average annual range from 400 to 800 mm, is not evenly distributed, which is more in the south and less in the north. In the study area, the area of natural grassland is 26 030 km^2, which takes up 67.64% of the total area, and the available grassland area is approximately 24 874 km^2, which accounts for 95.56% of the grassland area. Due to its unique geographical location and climatic conditions, the study area has a rich variety of grassland types, of which the main grassland types are alpine meadow and alpine steppe [37].

B. Field Data Collection

The field investigation mainly focused on the grassland growing seasons (June to October) from 2016 to 2019. To completely monitor the study area, permanent monitoring areas and random monitoring areas were established throughout the study area based on its spatial heterogeneity, grassland availability, and management pattern. There were four permanent monitoring areas (AZ, XC, YLJ, and GJ), which were arranged from southwest to northeast (Fig. 1). Five sample plots of 100 m × 100 m were laid out in each permanent monitoring area and five subplots were laid out (0.5 m × 0.5 m) in each sample plot during 2016–2017. To reduce the effect of spatial heterogeneity, the subplots were collected as many as possible to accurately monitor the sample plots. In 2018 and 2019, nine subplots were set up in each sample plot. In addition, three permanent monitoring areas (GH, GM, and DS) were added in 2019 compared to previous years. A total of 575 subplots were collected in permanent monitoring areas during 2016–2019. For the random monitoring areas, the sizes of the sample plots and subplots were consistent with those in the permanent monitoring areas. The distances between the sample plots were greater than 5 km. Two or three subplots were laid out in a sample plot with consistent vegetation communities and three or four subplots were set up with more complex vegetation, which resulted in a total of 297 subplots being collected in the random monitoring

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TABLE I
TIME OF FIELD INVESTIGATION AND NUMBER OF SUBPLOTS COLLECTED

Type	Year	June	July	Aug.	Sept.	Oct.	Subplots
Permanent monitoring	2016	-	-	-	-	15	15
Random monitoring		-	-	-	-	-	-
Permanent monitoring	2017	20	20	20	20	20	100
Random monitoring		-	-	105	-	-	105
Permanent monitoring	2018	36	36	36	36	36	180
Random monitoring		-	-	102	-	-	102
Permanent monitoring	2019	56	56	56	56	56	280
Random monitoring		-	-	90	-	-	90

areas (Table I). All in all, the NDF content was collected from 168 sample plots from June to October of 2016–2019 in the study area.

In each subplot, the canopy spectral reflectance was measured using a portable ASD Field Spec Pro FR2500 spectrometer (Analytical Spectral Devices Inc., Boulder, CO, USA) whose spectral bands were 350–2500 nm with sampling intervals of 1 nm. Sampling was conducted under weather conditions of no or few clouds, always between 10:00 and 14:00, and at a height of approximately 1 m from the sensor to the canopy. The measurement of each subplot was repeated ten times. In the spectral analysis, spectra with obvious anomalies were eliminated at first since spectral measurements were susceptible to factors such as weather and air moisture. In addition, the weather conditions at the time of the measurement and the observation time were recorded, as well as indicators such as the grassland types, dominant species, number of species, vegetation coverage, height, and geographic location information for each subplot. Furthermore, the above-ground parts of all of the vegetation were mowed, brought back to the laboratory, dried to a constant weight in a constant-temperature oven at 65 °C, weighed, crushed, sieved (1 mm), and taken for chemical analysis in the laboratory. The NDF content was determined by the ANKOM filter bag procedure with the ANKOM A200i fiber analyzer [35]. Each sample was weighed to be exactly 1.0 g and placed in a tall beaker, and a total of 100 mL neutral detergent solution, a few drops of decahydronaphthalene (an antifoamer), and 0.5 g anhydrous sodium sulfite were added to the beaker. Then, a condensation unit was attached, and the sample was immediately placed on an electric stove to boil as soon as possible (5–10 min). After the solution boiled, the electric stove was adjusted to keep it at a low boil (to prevent bubbles from rising). The beaker was shaken frequently to thoroughly mix and reflux the sample and the solution for 1 h. Then, after it was removed from the stove, the beaker was cooled for 10 min, and a constant-weight crucible was placed on a filter flask. The whole residue was transferred into the crucible, and the solution was filtered by vacuum until the filtrate became neutral (tested with blue litmus paper). Next, the residue was rinsed twice with 20 mL acetone until the flowing acetone solution was colorless. The crucible was moved into a 100 °C air-drying oven for 3 h, then transferred to a desiccator for 30 min of cooling and weighed until its weight was constant. In this study, the bag method was also used to analyze the contents of ADF, crude fiber (CF), cellulose, hemicellulose, and lignin, with the ADF analyzed based on an acidic detergent solution.

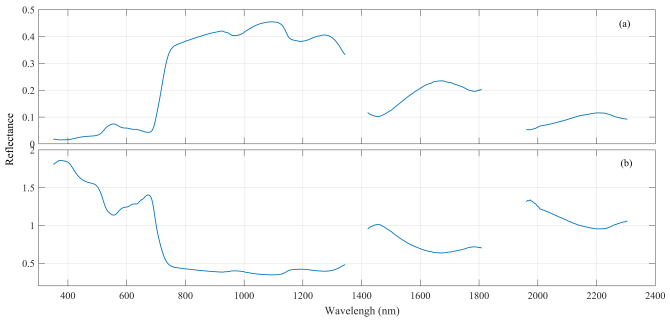


Fig. 2. (a) Mean OR and (b) mean Log spectra of 168 sample plots from 2016 to 2019.

C. Spectral Transformation Techniques

The smoothing and filtering of the spectral reflectance can effectively eliminate noise and improve the signal-to-noise ratios without changing the shape or width of the signal. Therefore, in this study, filtering was performed using the Savitzky-Golay filter [38]. The spectral bands at 1345–1420 nm, 1771–2018 nm, and 2306–2500 nm were found to be significantly impacted by the absorption of leaf water and were thus removed. Spectral transformations, including Log, FD, and CR, were then applied. These transformation methods can effectually decrease or eliminate the impacts of atmospheric factors, water absorption, and soil background [39], [40]. The CR and FD transformations were performed by ENVI 5.3 software. In addition, the Log spectra were obtained by taking the logarithm of the reciprocal of the original spectral reflectance (Fig. 2). The data of 1729 OR, FD, Log, and CR spectral bands (350–1302, 1473–1770, and 2019–2305 nm) were analyzed.

D. Spectral Absorption Feature Parameters

Table II summarizes the spectral absorption feature parameters, including the variables of the spectral position, spectral area, and VIs. Most of the listed absorption feature parameters have been previously successfully applied in research on fiber components [14], [21], [41], [42]. Some of the parameters are calculated using the hyperspectral sensitive bands selected in this study. Among them, SR is the ratio of the reflectance of every two sensitive bands, and the bands are 682, 683, 687, 727, 728, 2276, 2283, 2303, and 2305 nm. NDSI is the result based on a combination of the reflectance of two shortwave infrared (SWIR) bands, and NDLI is acquired by taking the logarithm of the reciprocal of the two SWIR bands. The spectral bands used by the two VIs are 2276, 2283, 2303, and 2305 nm.

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TABLE II
SPECTRAL ABSORPTION FEATURE PARAMETERS FOR NDF CONTENT ESTIMATION

Variable type	Variables	Formula and Description	References
Variables of spectral position	BD	Absorption depth, $BD = 1 - R'$, R' = continuum-removed spectral reflectance value	[21]
	BDC	Band depth of band center	
Variables of spectral area	SD_b	Summation of the first order derivatives in 490-530 nm	[43]
	SD_y	Summation of the first order derivatives in 550-582 nm	
	SD_r	Summation of the first order derivatives in 680-780 nm	
	SD_g	Summation of the first order derivatives in 510-560 nm	
	SR_r	Summation of the first order derivatives in 640-680 nm	
VIs	T1	SD_r/SD_b	[43]
	T2	SD_r/SD_y	
	T3	SD_r/SD_g	
	T4	SD_r/SR_r	
	T5	$(SD_r - SD_b)/(SD_r + SD_b)$	
	T6	$(SD_r - SD_y)/(SD_r + SD_y)$	
	T7	$(SD_r - SD_g)/(SD_r + SD_g)$	
	T8	$(SD_r - SR_r)/(SD_r + SR_r)$	
	SR01	SR depicts simple ratio index, $SR01 = \rho 800/\rho 670$	[44]
	NDSI	NDSI depicts normalized difference spectral indices, $NDSI = (\rho 1 - \rho 2)/(\rho 1 + \rho 2)$, ρ1 is 1222, 1236, 905, 1011, 1192, 1173 and 1204 nm, and ρ2 is 1281, 922, 1012, 1258, 1225, 1206 and 1179 nm, respectively, denoted by NDSI01-NDSI08	[42]
	NDLI01	NDLI depicts normalized difference lignin index, $NDLI01 = \frac{(\log(1/\rho 1754) - \log(1/\rho 1680))}{(\log(1/\rho 1754) + \log(1/\rho 1680))}$	[41]
	BDR	BDR depicts band depth ratio, $BDR = BD/BDC$	[21]
	NBDI	NBDI depicts normalized band depth index, $NBDI = (BD - BDC)/(BD + BDC)$	
	SR	$SR = \rho 1/\rho 2$, ρ is 682, 683, 687, 727, 728, 2276, 2283, 2303 and 2305 nm, denoted by SR1-SR36	They are calculated using the hyperspectral sensitive bands selected in this study
	NDSI	$NDSI = (\rho 1 - \rho 2)/(\rho 1 + \rho 2)$, ρ is 2276, 2283, 2303 and 2305 nm, denoted by NDSI1-NDSI6	
	NDLI	$NDLI = \frac{(\log(1/\rho 1) - \log(1/\rho 2))}{(\log(1/\rho 1) + \log(1/\rho 2))}$, ρ is 2276, 2283, 2303 and 2305 nm, denoted by NDLI1-NDLI6	

E. Environmental Factors

EFs, including mainly topography, soil, meteorology, and vegetation, affect the distribution of water and nutrients in soil and plants [45]–[47]. Among these, topography mainly includes the profile curvature, curvature, aspect, slope, plan curvature, and elevation. Digital elevation model (DEM), with a spatial resolution of 90 m, is derived from Shuttle Radar Topography Mission (SRTM) images (http://srtm.csi.cgiar.org/) and is resampled to 100 m. The aspect, slope, and solar radiation (SOL) are extracted in ArcGIS 10.2, and then, the plan curvature and profile curvature are calculated using the aspect and slope, respectively. Soil data (http://globalchange.bnu.edu.cn/research/soil) includes the clay contents of the surface (0–30 cm) (Clay1) and bottom (30–60 cm) (Clay2) and sand contents of the surface (Sand1) and bottom (Sand2) soil as well as the ratio of the two [48]. Meteorological data includes the temperature, precipitation, and SOL, and the temperature and precipitation are obtained from the National Earth System Science Data Center, National Science & Technology Infrastructure of China (http://www.geodata.cn). Vegetation data consists of the grass height measured in the field. The grassland cover is calculated by unmanned aerial vehicle (UAV) images. The diagonal of the sample plot (100 m × 100 m) was selected as the flight route, and the UAV recorded aerial images every 3 s at the height of 30 m, with a flight speed of 4 m/s. Approximately 12 photographs were taken in each sample plot, and the UAV only took approximately 2 min to complete a photography mission in a plot. The grassland cover of each photograph was extracted using the photograph analyzing module of FragMAP [49]. The covers of the 12 photographs were then averaged as the overall grassland cover of the plot.

F. Data Analysis and Modeling

1) Selection of Variables: In our study, the least absolute shrinkage and selection operator (LASSO) regression is used to select the sensitive variables. This approach can significantly decrease the number of variables to achieve dimensionality reduction. LASSO regression is based on the principle of the least-squares method. The regression coefficient of each variable in the model is used to construct a certain function as a "penalty function," and the coefficient of each variable is compressed until the residual sum of squares of the model is minimized. The variables whose coefficients are reduced to 0 for the special penalty function are eliminated [50], [51]. For this reason, the LASSO regression method is widely used to select variables from high-dimensional data such as hyperspectral data [52]–[54]. In our study, the glmnet package (version 3.0-2) in R software (version 3.6.3) was used to accomplish the LASSO regression.

2) RF Algorithm: Proposed by Cortes and Vapnik [57], RF is based on many decision trees to enhance the classification and regression tree (CART). In this algorithm, trees are constructed independently of each other through random sampling with replacement. Two main parameters need to be set.

1) ntree, which is the number of regression trees constructed and is based on the number of samples. The higher the value of ntree, the more stable the result of the dependent variables in the simulation is [56]. Nevertheless, setting ntree too high will increase the computational burden.
2) mtry, which is the number of predictors tested at each node [33]. The default value of 500 is used for ntree and mtry is automatically determined as the square root of the total number of predictors used. In this study, the RF algorithm is implemented using the randomForest package (version 4.6-14).

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3) SVM Algorithm: SVM is based on Vapnik–Chervonenkis (VC) dimension theory and targets small-sample data. The core idea of the algorithm is to map low-dimensional vectors to high-dimensional feature space to construct a perfect "hyperplane" for classification according to certain non-linear mapping rules (the kernel function) [57]. The kernel functions include radial basis, linear, polynomial, and sigmoid functions [58]. The SVM model is built using the e1071 package (version 1.7-3) in R software (version 3.6.3). Because of the nonlinearity of the regression problem that has to be solved in our study, the radial basis function is selected as the kernel function.

4) ANN Algorithm: ANN is an analysis method mimicking the working mechanism of the human brain, and there is a wide variety of related algorithms. Among them, the back-propagation ANN (BP-ANN) is used in diverse fields due to its robustness and flexible control system [59]–[61]. It includes three layers of neurons, namely, the input layer, the hidden layer, and the output layer, and the neurons in different layers are independent of each other. When the algorithm starts, several signals in the input layer are weighted and then passed to the hidden layer where they are processed by the activation function before being passed to the output layer. If the output value satisfies the limit of the error function, then the result is output; otherwise, the weight of each individual signal continues to be corrected until the error is minimized [62], [63]. The BP-ANN model is constructed using the newff function of the AMORE package (version 0.2-16) in R software (version 3.6.3).

G. Accuracy Evaluation

The method of tenfold cross-validation assesses the accuracy and stability of the models [64], [65]. Before the establishment of the model, the dataset is randomly divided into ten groups and per group has an equal number of data. For each cross-validation run, one group is selected as a test dataset to validate the estimation ability of the model, and the remaining nine groups are used as training sets to build the model. Each time the cross-validation is performed, the R^2 and RMSE between the predicted and measured values are calculated. This process is conducted ten times, and finally, the average values of R^2 and RMSE are used as indicators for estimating the performance of the model. The calculation is as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2} \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2}{n}} \tag{2}$$

where $\hat{Y}_i$, $\bar{Y}$, and Y_i are the predicted, average, and measured values, respectively, of the test set, and n is the number of data. A larger value of R^2 indicates higher predictive reliability of the model, while a smaller RMSE signifies that the model better fits the data [66].

TABLE III

DESCRIPTIVE STATISTICS OF THE FORAGE NDF CONTENT IN THE STUDY AREA FROM 2016 TO 2019

Years	Sample plots	Min (%)	Max (%)	Mean (%)	SD (%)	CV
2016	3	43.43	60.02	49.89	4.45	0.09
2017	56	34.47	67.94	49.39	6.27	0.13
2018	51	38.68	65.47	50.73	6.13	0.12
2019	58	29.35	69.41	51.41	6.93	0.13
All	168	29.35	69.41	50.66	6.52	0.13

TABLE IV

CORRELATIONS BETWEEN THE FIBER COMPONENTS OF FORAGE IN NATURAL ALPINE GRASSLAND

Fiber Components	CF	NDF	ADF	Hemicellulose	Cellulose	Lignin
CF	1					
NDF	0.69**	1				
ADF	0.62**	0.75**	1			
Hemicellulose	0.50**	0.84**	0.28**	1		
Cellulose	0.62**	0.85**	0.84**	0.56**	1	
Lignin	0.18*	-0.01	0.42**	-0.36**	-0.08	1

* means $P < 0.05$, and ** means $P < 0.01$.

CF: crude fiber; NDF: neutral detergent fiber; ADF: acid detergent fiber.

III. RESULTS

A. Descriptive Statistics of the NDF Content

The statistical results of the NDF of 168 sample plots collected from June to October of 2016–2019 in the study area showed that, during the four-year period, the average NDF content ranged from 49.39% to 51.41%. The coefficient of variation (CV) of the NDF ranged from 0.09 to 0.13. The CV was higher from 2017 to 2019 than that in 2016 (Table III). This finding could be related to spatial and temporal differences. In particular, the samples collected in the field from 2017 to 2019 were more dispersed and numerous, and the samples were collected at multiple different growth stages.

There is a complex relationship among the various fiber components. We analyze the correlations between the contents of NDF and the other fiber components in forage. The correlations are extremely significant between the contents of the other five fiber components except for lignin, but not significant between the contents of lignin and some fiber components ($P < 0.01$). Among the other five fiber components, except for the relatively low correlation between ADF and hemicellulose ($r = 0.28$), the correlation coefficients between the contents of any two fiber components are above 0.50. Among them, the correlation coefficients between the NDF and the other four fiber components range from 0.69 to 0.85, and the correlation coefficient between the NDF and cellulose is the highest, reaching 0.85, followed by that between the NDF and hemicellulose content, which is 0.84 (Table IV). This finding indicates that there is a complex relationship among the five fiber components besides lignin, and NDF content is significantly correlated with those of the other four fiber components.

B. Correlation Analysis of the Forage NDF With Spectral Bands

Correlation analyses are performed to examine the relationship between forage NDF and the four transformations

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TABLE V
COMPARISON OF SENSITIVE BANDS SELECTED BASED ON CORRELATION ANALYSIS AND THE LASSO METHOD

The transformed spectra	Methods	Number of bands	Sensitive bands (nm)		
			Visible	NIR	SWIR
OR	Correlation analysis ($P < 0.05$)	1458	350-479, 505-593, 699-780	781-1344	1474-1809, 1964-1966, 2052-2305
	LASSO	9	682, 683, 687, 727, 728	-	2276, 2283, 2303, 2305
Log	Correlation analysis ($P < 0.05$)	1625	350:607, 697-780	781-1344	1425:1441,1446:1809,1961:2027,2030:2305
	LASSO	10	374, 675, 678, 681, 727, 728, 729	-	2276, 2283, 2303
FD	Correlation analysis ($P < 0.05$)	750	350-351, 358-363, 365-366, 374-376, 385-386, 391, 397-399, 405, 416-417, 420-425, 428-429, 437, 444-445, 460-472, 474-476, 478-482, 485-486, 496-549, 554-664, 666-670, 680-739, 768-780	781-811, 816-851, 855-889, 892-894, 898-900, 904-905, 908-909, 912-916, 919-923, 938-939, 942-946, 949, 952-959, 962-963, 967-968, 975-979, 988, 998-999, 1002-1005, 1036, 1047-1059, 1064-1070, 1082-1084, 1125, 1127-1128, 1132-1133, 1153, 1157, 1165-1166, 1176-1180, 1184-1187, 1191, 1195, 1224-1244, 1248-1250, 1268-1281, 1283-1287, 1289-1290, 1295-1298, 1301, 1334	1461-1462, 1474, 1482-1550, 1552-1575, 1579-1600, 1602-1606, 1621-1623, 1627-1628, 1650, 1665-1666, 1698-1699, 1714, 1720-1724, 1731-1752, 1755-1764, 1768, 1769, 1770, 1773, 1774, 1775, 1779, 1785-1786, 2030-2031, 2035, 2050-2053, 2078-2079, 2080-2082, 2107-2109, 2125-2127, 2152, 2178, 2185-2186, 2214-2215, 2222-2224, 2240-2246, 2268, 2291
	LASSO	18	624, 630, 704	840-841, 1004, 1083, 1178, 1271	1449, 1485, 1522, 1546, 1574, 1744, 2091, 2107, 2246
CR	Correlation analysis ($P < 0.05$)	84	652-694, 742-769	1113-1117	2234-2241
	LASSO	18	688-694, 745-746, 761	1057, 1088, 1116, 1287	2111, 2116, 2242, 2276

OR spectra: original spectra; Log spectra: log transformation spectra; FD spectra: first derivative spectra; CR spectra: continuum removal spectra.

of the spectral data (OR, Log ($1/R$), FD, and CR). The sensitive bands of different spectra are selected, as shown in Table V. According to the significance test, for the OR spectra, 1458 bands are significantly correlated with NDF content ($|r| > 0.150$, $P < 0.05$), which are mainly in the blue (350–479, 505–520 nm), green (521–593 nm), red-edge (699–780 nm), NIR (781–1344 nm), and SWIR (1474–1809, 1964–1966, 2052–2305 nm) regions. The correlation between the band at 723 nm and NDF content is the highest ($r = -0.43$) [Fig. 3(a)].

The relationship between the Log spectra and the NDF is shown in Fig. 3(b). There are 1625 significantly correlated bands ($|r| > 0.150$, $P < 0.05$), whose correlation coefficients exhibit trends similar to those of the OR. The range of significantly correlated bands is also similar, and the correlation between the band at 718 nm and NDF content is the highest ($r = 0.43$).

The correlation between the FD spectral reflectance and the NDF is shown in Fig. 3(c). There are 750 significantly correlated bands ($|r| > 0.150$, $P < 0.05$), which span over a relatively broad range, including the blue, green, red, red-edge, NIR, and SWIR regions. The correlation between the band at 1522 nm in the SWIR region and NDF content is the highest ($r = -0.47$).

There are relatively few bands in which the CR spectral reflectance correlated with the NDF. There are 84 significantly correlated bands ($|r| > 0.150$, $P < 0.05$), which are concentrated in the red (652–690 nm), red-edge (691–694 and 742–769 nm), NIR (1113–1117 nm), and SWIR (2234–2241 nm) regions. The correlation between the band at 760 nm and NDF is the highest ($r = -0.26$) [Fig. 3(d)].

C. Results of the LASSO Regression

The LASSO regression analysis is used to select important variables that exert great influence on forage NDF. The number of selected variables of the OR and three transformed spectra, VIs and EFs ranges from 5 to 18. The bands are in the red (682, 683, 687 nm, denoted O_682, O_683, O_687, the same below), red-edge (O_727, O_728), and SWIR regions (O_2276, O_2283, O_2303, O_2305) selected from the OR spectral data (Table V). Compared with the sensitive bands selected by the LASSO method, the same bands selected by correlation analysis are in the red and SWIR regions, the different bands are in the blue, green, and NIR regions. The band selected from the Log spectra includes a new blue region (L_374) which is not selected from OR spectra, but other selected bands are similar to those from the OR spectra. The bands are in the red (F_624, F_630), red-edge (F_704), NIR (F_840, F_841, F_1004, F_1083, F_1178, F_1271), and SWIR (F_1449, F_1485, F_1522, F_1546, F_1574, F_1744, F_2091, F_2107, F_2246) regions, while the sensitive bands selected by correlation analysis cover a wide range (from the blue to SWIR regions) selected from the FD spectral data. The bands selected from the CR spectra are in the red, red-edge, NIR, and SWIR regions, and the range is consistent with the sensitive bands selected by correlation analysis. In addition, among the EFs, the plan curvature, precipitation, Clay1, height, and SOL strongly influence the NDF, while the SR16, SR18, NDSI01, BDC, and SDg of VIs substantially affect the NDF content.

The bands selected by the correlation analysis mostly cover those selected by the LASSO method. Compared with the correlation analysis method, the LASSO method substantially reduces the number of variables and achieves significant

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Fig. 3. Correlations between (a) NDF content and OR, (b) Log spectra, (c) FD spectra, and (d) CR spectra.

dimension reduction. Therefore, the LASSO method is used in our study to select important variables from four different data types (OR, Log, FD, and CR) and from the VIs and EFs. The six types of data described above are combined as shown in Table VI (23 categories), and then are selected again by the LASSO method. Among the total 29 data types, the number of variables selected by LASSO ranges from 5 to 28 (Table VI).

D. NDF Models and Accuracy Evaluation

RF, SVM, and ANN models are constructed based on the spectral absorption feature parameters, the VIs, and the EFs, resulting in a total of 87 NDF models. The results of the tenfold cross-validation are shown in Table VII. In the training set, the models have R^2 between 0.35 and 0.95 and RMSE between 1.74% and 4.89%. In the validation set, the models have R^2 between 0.22 and 0.62 and RMSE between 3.95% and 5.03%. Evidently, the R^2 is higher for the training set than for the validation set, while the RMSE is lower for the training set than for the validation set.

In terms of the 12 models constructed based on every type of four spectral data using RF, SVM and ANN methods, the RF model based on the FD spectra ($R^2 = 0.51$, RMSE = 4.31%) is more precise than the models constructed based on the other three spectral data. For the 12 models constructed by combining every type of four spectral data with VIs, the RF models based on the FD-VIs ($R^2 = 0.59$, RMSE = 3.96%) have higher accuracy. Moreover, for the 12 models constructed by combining every type of four spectral data with EFs, the FD-EFs-RF ($R^2 = 0.55$, RMSE = 4.14%) model has higher accuracy. The RF model based on FD-VIs-EFs gives the best results ($R^2 = 0.61$, RMSE = 4.05%) when the VIs and EFs are all combined with every type of four spectral data (12 models). The combination of VIs and EFs can enhance the accuracy of the NDF models. Therefore, in further study, we construct the models using different types of spectral data combined with VIs and EFs all the way, then compare and analyze the performance of different models on the estimation of NDF content.

As seen from the 18 models constructed using every two types of spectral data (OR, FD, Log, and CR) combined with VIs and EFs, the RF model based on FD-Log-VIs-EFs features the highest estimation accuracy, with R^2 of 0.62 and RMSE of 3.98%. Among the 12 models constructed using every three types of spectral data (OR, FD, Log, and CR) combined with VIs and EFs, the R^2 of the RF model based on OR-CR-FD-VIs-EFs reaches 0.60, while the RF model exhibits the highest accuracy (OR-CR-FD-Log-VIs-EFs)

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TABLE VI
LASSO REGRESSION ANALYSIS RESULTS

The types of data	Number of variables before selecting	Number of variables after selecting	The types of data	Number of variables before selecting	Number of variables after selecting
OR	1735	9	Log-VIs-EFs	20	10
Log	1735	10	FD-VIs-EFs	28	28
FD	1735	18	CR-VIs-EFs	28	10
CR	1735	18	OR-CR-VIs-EFs	37	10
VIs	74	5	OR-FD-VIs-EFs	37	19
EFs	17	5	OR-Log-VIs-EFs	29	8
OR-VIs	14	5	CR-FD-VIs-EFs	46	19
Log-VIs	15	5	CR-Log-VIs-EFs	38	11
FD-VIs	23	16	FD-Log-VIs-EFs	38	22
CR-VIs	23	11	OR-CR-FD-VIs-EFs	55	15
OR-EFs	14	4	OR-CR-Log-VIs-EFs	47	10
Log-EFs	15	4	OR-FD-Log-VIs-EFs	47	19
FD-EFs	23	19	CR-FD-Log-VIs-EFs	56	20
CR-EFs	23	19	OR-CR-FD-Log-VIs-EFs	65	20
OR-VIs-EFs	19	11			

The important variables selected from VIs are SR16, SR18, NDSI01, BDC and SDg, and selected from EFs are Plan_curvature, SOL, Precipitation, Clay1 and Height, and selected from FD-Log-VIs-EFs are F_704, F_840, F_841, F_1004, F_1083, F_1271, F_1449, F_1485, F_1522, F_1744, F_2091, F_2107, F_2246, SR16, SR18, NDSI01, Plan_curvature, SOL, Precipitation, Clay1, Height and L_2276. O_** represents the bands selected from the OR spectra; L_** represents the bands selected from the Log spectra; F_** represents the bands selected from the FD spectra; C_** represents the bands selected from the CR spectra.

(R^2 = 0.59, RMSE = 4.01%) among the models constructed using the four types of spectra (OR, FD, Log, and CR) combined with VIs and EFs.

Overall, the RF model based on FD-Log-VIs-EFs has the highest accuracy, with the R^2 and RMSE of the validation set reaching 0.62 and 3.98%, respectively, followed by the RF models based on FD-VIs-EFs, OR-FD-VIs-EFs, and CR-FD-VIs-EFs, each with R^2 values of 0.61. Of the three machine learning models based on these different data types, the RF models are more accurate than the other two models (SVM and ANN) in most cases. However, of the models based on the CR and CR-EFs, the ANN models have the highest accuracy. In addition, the models constructed by combining spectral data of different types with VIs and EFs generally have significantly higher accuracy than the models constructed by only spectral data, which suggests that the combination of important spectral bands with VIs and EFs can increase the accuracy of NDF content estimation.

To further analyze the influence of VIs and EFs on the forage NDF content, the accuracy of different NDF models is evaluated by the tenfold cross-validation and Taylor diagram methods. The statistical results of the 48 estimation models are shown in Fig. 4. The RF models (red icons in Fig. 4) are relatively better than the SVM and ANN models. In addition, the models based on spectral bands combining VIs and EFs exhibit better results. Among them, the models constructed using FD-VIs-EFs-RF (R^2 of 0.61 and RMSE of 4.05%) and FD-VIs-RF (R^2 of 0.59 and RMSE of 3.96%) performed better than the other models because the standard deviations of the models are lower and predicted points were closer to the observed points (Fig. 4).

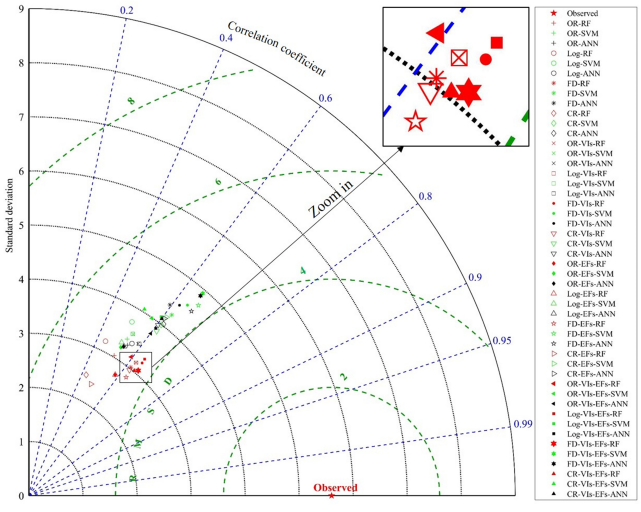


Fig. 4. Taylor diagram showing a statistical comparison of the 48 forage NDF models.

E. Evaluation of the Influence of Variables on NDF Content

Overall, the combination of sensitive spectral bands based on different data types with important EFs and VIs variables can increase the evaluation accuracy of the forage NDF content. Among the 87 models constructed in the study, the best

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TABLE VII
COMPARISON OF THE ACCURACY OF DIFFERENT NDF MODELS

The types of data	Number of variables	Models	Training set		Validation set	
			R^2	RMSE (%)	R^2	RMSE (%)
OR	9	RF	0.91	2.18	0.46	4.58
		SVM	0.47	4.10	0.35	4.59
		ANN	0.35	4.51	0.35	4.61
Log	13	RF	0.91	2.19	0.43	4.87
		SVM	0.48	4.06	0.36	4.62
		ANN	0.36	4.48	0.35	4.58
FD	18	RF	0.94	1.89	0.51	4.31
		SVM	0.49	3.99	0.48	4.31
		ANN	0.52	3.89	0.44	4.61
CR	18	RF	0.93	2.18	0.40	4.92
		SVM	0.47	4.08	0.40	4.39
		ANN	0.49	4.01	0.45	4.39
VIs	5	RF	0.90	2.20	0.50	4.31
		SVM	0.53	3.87	0.36	4.66
		ANN	0.37	4.45	0.35	4.69
EFs	5	RF	0.91	2.54	0.45	4.82
		SVM	0.40	4.40	0.29	5.03
		ANN	0.24	4.89	0.22	5.02
OR-VIs	5	RF	0.90	2.21	0.51	4.28
		SVM	0.55	3.80	0.41	4.63
		ANN	0.38	4.41	0.37	4.54
Log-VIs	5	RF	0.90	2.21	0.51	4.28
		SVM	0.54	3.81	0.40	4.51
		ANN	0.38	4.40	0.37	4.53
FD-VIs	16	RF	0.94	1.80	0.59	3.96
		SVM	0.53	3.85	0.50	4.31
		ANN	0.55	3.74	0.47	4.42
CR-VIs	11	RF	0.93	1.97	0.52	4.24
		SVM	0.46	4.13	0.44	4.28
		ANN	0.47	4.06	0.45	4.32
OR-EFs	4	RF	0.90	2.36	0.49	4.35
		SVM	0.55	3.93	0.36	4.74
		ANN	0.38	4.56	0.37	4.64
Log-EFs	4	RF	0.90	2.36	0.49	4.43
		SVM	0.56	3.85	0.37	4.85
		ANN	0.38	4.56	0.37	4.63
FD-EFs	19	RF	0.95	1.83	0.55	4.14
		SVM	0.57	3.66	0.52	4.07
		ANN	0.60	3.53	0.48	4.23
CR-EFs	19	RF	0.95	2.07	0.44	4.76
		SVM	0.51	3.92	0.44	4.43
		ANN	0.57	3.67	0.46	4.32
OR-VIs-EFs	11	RF	0.93	1.90	0.52	4.93
		SVM	0.72	3.11	0.41	4.53
		ANN	0.46	4.13	0.40	4.45
Log-VIs-EFs	10	RF	0.93	1.91	0.54	4.09
		SVM	0.48	4.07	0.41	4.50
		ANN	0.46	4.13	0.43	4.46
FD-VIs-EFs	**13**	**RF**	**0.95**	**1.74**	**0.61**	**4.05**
		SVM	0.64	3.36	0.52	4.23
		ANN	0.68	3.19	0.49	4.35
CR-VIs-EFs	10	RF	0.94	1.91	0.55	4.23
		SVM	0.75	2.97	0.45	4.40
		ANN	0.48	4.03	0.47	4.49
OR-CR-VIs-EFs	10	RF	0.93	1.91	0.53	4.13
		SVM	0.76	2.88	0.48	4.19
		ANN	0.49	4.01	0.45	4.30
OR-FD-VIs-EFs	**19**	**RF**	**0.95**	**1.75**	**0.61**	**4.06**
		SVM	0.60	3.53	0.53	4.01
		ANN	0.63	3.39	0.49	4.25
OR-Log-VIs-EFs	8	RF	0.93	1.97	0.55	4.11
		SVM	0.70	3.19	0.42	4.48
		ANN	0.45	4.14	0.42	4.40
CR-FD-VIs-EFs	**19**	**RF**	**0.95**	**1.74**	**0.61**	**4.04**
		SVM	0.70	3.19	0.42	4.48
		ANN	0.63	3.40	0.51	4.22
CR-Log-VIs-EFs	11	RF	0.93	1.89	0.54	4.21
		SVM	0.73	2.88	0.46	4.32
		ANN	0.48	4.01	0.43	4.46
FD-Log-VIs-EFs	**22**	**RF**	**0.95**	**1.74**	**0.62**	**3.98**
		SVM	0.86	2.24	0.53	4.06
		ANN	0.65	3.33	0.52	4.23
OR-CR-FD-VIs-EFs	15	RF	0.95	1.74	0.60	4.08
		SVM	0.57	3.70	0.52	4.14
		ANN	0.60	3.54	0.50	4.33
OR-CR-Log-VIs-EFs	10	RF	0.93	1.92	0.52	4.09
		SVM	0.76	2.91	0.49	4.31
		ANN	0.49	4.01	0.45	4.30
OR-FD-Log-VIs-EFs	19	RF	0.95	1.74	0.58	4.00
		SVM	0.60	3.52	0.54	3.95
		ANN	0.63	3.40	0.51	4.23
CR-FD-Log-VIs-EFs	20	RF	0.95	1.74	0.58	3.96
		SVM	0.60	3.56	0.51	4.12
		ANN	0.63	3.41	0.51	4.28
OR-CR-FD-Log-VIs-EFs	20	RF	0.95	1.75	0.59	4.01
		SVM	0.60	3.52	0.54	4.00
		ANN	0.63	3.39	0.51	4.27

OR spectra: original spectra; Log spectra: log transformation spectra; FD spectra: first derivative spectra; CR spectra: continuum removal spectra; VIs: vegetation indices; EFs: environmental factors.

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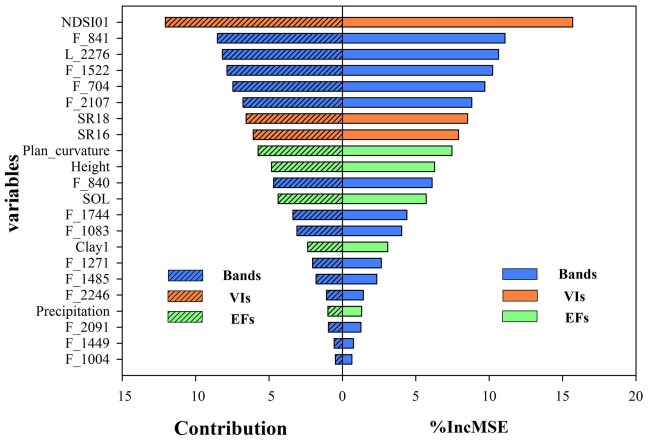


Fig. 5. Importance and contribution of each variable in the best model (FD-Log-VIs-EFs-RF) for prediction of the NDF. (L_** represents the bands selected from the Log spectra; F_** represents the bands selected from the FD spectra.).

model (FD-Log-VIs-EFs-RF) for estimating the NDF involved 22 variables. Among these variables, NDSI01, F_841, and L_2276 are more important for estimating the NDF content, and NDSI01 has the highest contribution (12.06%), followed by the reflectance of the FD spectra at 841 nm (8.52%) and the reflectance of Log spectra at 2276 nm (8.18%), while the reflectance of the FD spectra at 1004 nm has the lowest contribution (see Fig. 5).

The partial dependence plots (PDPs) of each variable in the model can show the quantitative effect of each variable on the NDF content. These plots indicate that the NDF responded nonlinearly to each variable. Of the 14 sensitive bands, the bands at 840, 841, 1271, and 1744 nm impacted NDF positively, and other bands have negative impacts. The band at 841 nm has the largest positive impact, with the NDF content increasing from 48.0% to 49.5% as reflectance rises from 0.03 to 0.05. The band at 1522 nm has the largest negative impact, with the NDF content decreasing from 49.2% to 48.0% as reflectance rises from 0.08 to 0.12. Among the EFs, Clay1 has little impact on the NDF, changing the content by only 0.3%. The grass height has a positive impact, causing a rise of 1.2% in the NDF content. The plan curvature, precipitation, and SOL have negative impacts, leading to decreases of 1.0%, 0.8%, and 0.6%, respectively, on the NDF content. Among the VIs, the NDSI01, SR16, and SR18 all have positive impacts on the NDF content, and the NDSI01 has the largest positive impact, with an increase of 2.5% in the NDF content as the value of NDSI01 rises from 0.02 to 0.03. Overall, NDSI01 and the band at 1522 nm have the largest positive and negative impacts, respectively, on the NDF content (Fig. 6).

IV. Discussion

A. Selection of Sensitive Bands by the LASSO Method

In this study, the spectral bands, the VIs, and EFs are selected by the LASSO regression method. The selection results show that the numbers of variables from the four spectral data types (OR, FD, Log, CR spectra) are reduced by 99%, and the numbers of variables for VIs and EFs are reduced by 95% and 71%, respectively. The selected numbers of OR, FD, Log, and CR spectra (their spectral bands total 1729) are 9, 18, 10, and 18, respectively. Five important variables are each selected from 94 VIs and 17 EFs. In comparison, the numbers of OR, FD, Log, and CR sensitive spectra selected by correlation analysis are 1458, 1331, 750, and 84, respectively. A comparison of the two selection methods demonstrates that the LASSO method has a significant dimensionality-reducing effect on the hyperspectral remote sensing data, which is of great significance for the selection of variables for high-accuracy inversion of fiber components influenced by multiple sources and multiple factors. The selection of variables by LASSO regression depends on the penalty factor λ. The number of selected variables decreases as the value of λ increases, but an excessive λ value causes an increase in the model error [67]. Therefore, it is extremely important to determine a reasonable value of λ. In the study, tenfold cross-validation is used to determine the $\lambda 0$ (0.13) corresponding to the smallest RMSE of the best model (FD-Log-VIs-EFs-RF), and $\lambda 1$ (0.37) is taken at a standard deviation away from $\lambda 0$ as the final penalty factor (Fig. 7). In this case, the accuracy of the model is not significantly improved by reducing the number of variables, and overfitting of the model is thus avoided.

B. Spectral Transformation Techniques

Spectral transformation techniques play an important part in the high-accuracy inversion of the forage nutrient contents. FD transformation of spectra, which is a commonly used spectral enhancement method, is very sensitive to the spectral signal-to-noise ratios and can remove a variety of background and noise [68], [69]. These transformation techniques have been widely applied to the inversion of forage fiber components. For example, Pullanagari *et al.* [70] collected samples from four commercial farms in New Zealand (white clover and perennial ryegrass) and divided the dataset into the validation and training set at a ratio of 1:1. Using Microsoft Visual Basic software, they transformed the number of bands from 1900 bands/nm (1900/1 nm) to 380 bands/5 nm (380/5 nm). Based on FD-transformed spectra, the ADF, NDF, and lignin contents were estimated using the PLSR method, with R^2 values of the validation set reaching 0.82, 0.75, and 0.71, respectively. In our study, the OR spectra are transformed using the Log, FD, and CR methods. As seen from the accuracy of the non-parametric models constructed by combining the four spectral data with VIs and EFs (see Table VII), compared with the R^2 (0.35–0.46) and RMSE (4.58%–4.61%) of the model based on OR spectra, the R^2 of the inversion model based on FD spectra increases by 0.02–0.16, and the RMSE decreases by 0.27%–0.3%. For the models constructed with a combination of two data types, compared with the model based on OR-VIs (R^2 of 0.37–0.51 and RMSE of 4.28%–4.63%), the R^2 and RMSE of the model based on FD-VI increases by 0.08–0.18 and decreases by 0.12%–0.67%, respectively, and compared with those of the model based on OR-EFs (R^2 of 0.36–0.49 and RMSE of 4.35%–4.74%), the R^2 and RMSE of the model based on FD-EFs increases

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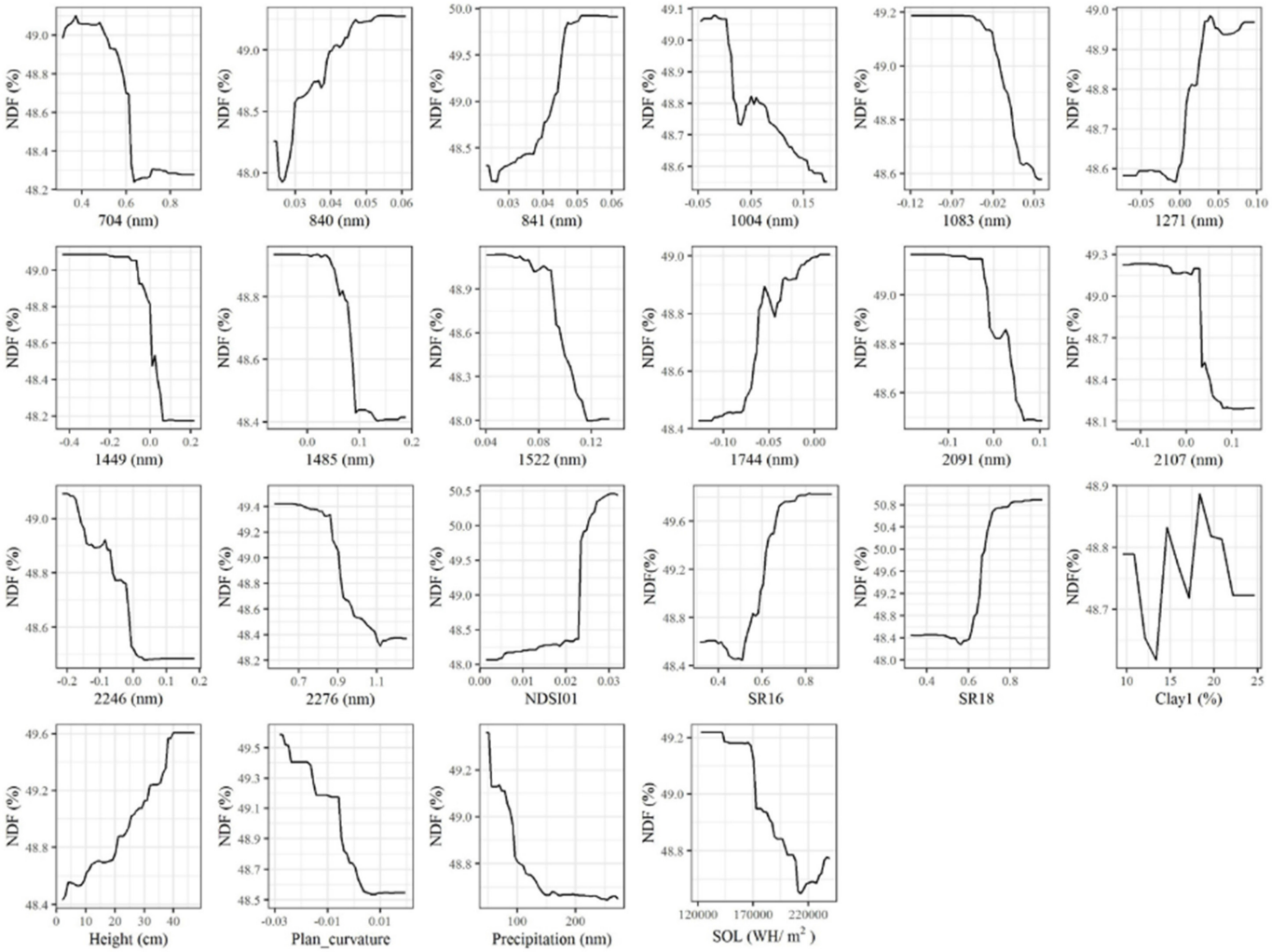


Fig. 6. PDPs of variables in the RF model based on FD-Log-VIs-EFs.

by 0.03–0.19 and decreases by 0.12%–0.67%, respectively. In the models constructed using a combination of three data types, compared with the model based on OR-VIs-EFs (R^2 of 0.40–0.52 and RMSE of 4.45%–4.93%), the R^2 and RMSE of the model based on FD-VIs-EFs increases by 0.03–0.21 and decreases by 0.10%–0.88%, respectively.

In addition, an analysis of the 87 models constructed in the study reveals that the four models (FD-Log-VIs-EFs-RF, CR-FD-VIs-EFs-RF, OR-FD-VIs-EFs-RF, and FD-VIs-EFs-RF) with relatively good results all have the variables of VIs, EFs, and FD spectra. For these four models, R^2 is between 0.61 and 0.62, and RMSE ranges from 3.98% to 4.06%. These results indicate that the FD spectra can indeed better reduce the noise and highlight more useful information and, thereby, significantly improve the inversion accuracy of the NDF content. The results also show that the VIs and EFs strongly influence the inversion of the NDF content.

On the other hand, the R^2 and RMSE of the training set of the four models described above reach 0.95 and range from 1.74% to 1.75%, respectively (Table VII). The difference in R^2 values between the training set and the validation set is approximately 0.34. This difference is somewhat high, perhaps because of the overfitting of the NDF model. We analyze the reasons in this study and find that the small sample size may be one of the reasons for RF overfitting [77].

C. Analysis of the Main Variables That Affect the Inversion Accuracy of the Forage NDF Content

1) Contribution of Spectral Variables to the Inversion of the Forage NDF Content: Our results show that the forage NDF content of alpine grassland is well correlated with CF, ADF, hemicellulose, and cellulose, but there are also complex relationships between these fiber components (Table IV). The NDF content includes cellulose, hemicellulose, lignin, and silicates. Cellulose is a major component of plant cell walls and is usually bound to hemicellulose, pectin, and lignin. The ADF content includes cellulose and acidic cellulose. Spectral reflectance and canopy structure of grasslands vary significantly in different growth stages, and nutrient contents vary considerably between different forage species. During the early growth stage, the protein content is high but the NDF content is low. As the plant grows, the digestible crude protein

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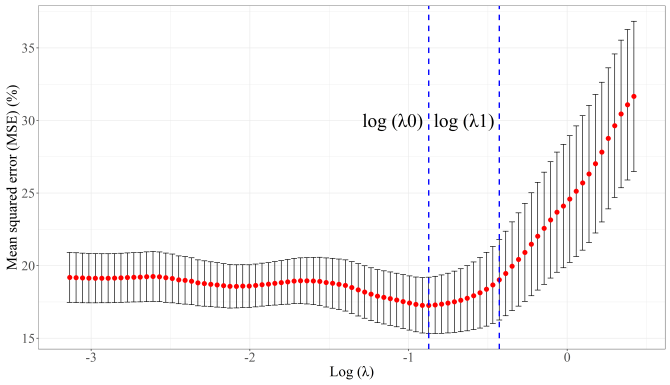


Fig. 7. Determination of the LASSO penalty factor for FD-Log-VIs-EFs-RF-based model using tenfold cross-validation: $\lambda 0$ is the penalty factor corresponding to the smallest RMSE of the model and $\lambda 1$ is the final penalty factor.

content gradually decreases, while the NDF content gradually increases [71], [72].

The spectral variables used in the best model (FD-Log-VIs-EFs-RF) for estimating the NDF content in the study mainly include the FD spectral channel and the Log spectral channel, which are mainly concentrated in the red-edge (F_704), NIR (F_840, F_841, F_1004, F_1083, F_1271), and SWIR (F_1449, F_1485, F_1522, F_1744, F_2091, F_2107, F_2246, L_2276) regions. Fig. 5 shows that the contributions of these bands range from 0.49% to 8.52%, the band at 841 nm contributing the most (8.52%), followed by the bands of 2276 and 1522 nm with contributions of 8.18% and 7.87%, respectively. In addition, the VIs used in the model are SR16, SR18, and NDSI01, whose contributions range from 6.09% to 12.06%, with NDSI01 contributing the most (12.06%), followed by SR18 (6.56%) and SR16 (6.09%) (Fig. 5). The OR spectral bands involved in these VIs are 687, 1222, 1281, 2276, and 2303 nm. Based on the relationship between the biochemistry and the wavelengths from the studies by Curran [73], Kokaly and Clark [21], and Kokaly *et al.* [74], among the 18 bands used in our best model, eight bands (1004, 1449, 1744, 2091, 2107, 2246, 2276, and 2303 nm) are similar to the known feature absorption bands of fiber components (cellulose, lignin, ADF, and NDF), and three bands (687, 704, and 1522 nm) are similar to the known feature absorption bands of nitrogen. Among them, the bands at 687 and 704 nm are in the red-edge regions. The band at 1004 nm contributes to the detection of NDF and plant starch contents, as it is susceptive to O-H stretch, and the bands at 1449, 1744, 2276 nm, which significantly impact the lignin and starch contents, are relevant to vibration mechanisms (i.e., O-H stretch, C-H stretch, and first overtone). The band at 1522 nm contributes to the detection of N content, as it is sensitive to the N-H bond, and the starch, cellulose, and the NDF content is related to the O-H deformation, CO stretching, C-H stretching, and C-C stretching at 2091 and 2107 nm. The bands at 2246 and 2303 nm contribute to the detection of protein, NDF, and ADF contents, as they are susceptible to C-H stretching, N-H stretching, and C = O stretching. It follows that some of the selected bands are associated with the known feature absorption bands of fiber components, but some bands (687, 704, 1522, and 2246 nm) are also associated with the known feature absorption bands of nitrogen, protein, and starch. This finding indicates that the prediction of alpine grassland forage NDF content on the Tibetan Plateau is correlated with both the known feature absorption bands of fiber components and those of other nutrient components.

Few studies have been conducted on the inversion of the NDF content based on the sensitive spectral bands of the original and three transformed spectra in combination of VIs and EFs, and some of the relevant studies have focused on the inversion of NDF by using the OR and FD spectra. For example, Zhao *et al.* [18] estimated the NDF using MAXR multiple regression and found that the prediction accuracy could be improved by combining a total of ten wavebands, which included the red (635, 645, and 675 nm), NIR (805, 823, 895, 1155, and 1165 nm), and SWIR (1635 and 1695 nm) regions. Zeng and Chen [25] found that the suitable bands for estimating the NDF in the mixed grassland (dominated by intermediate wheatgrass, pubescent wheatgrass, and alfalfa) were in the green (556, 560, 566, 570, 571, and 575 nm), red (644 nm), and NIR (1142 and 1145 nm) regions. Findings from both studies suggested that the red and NIR regions were important for estimating NDF content, which was consistent with our results. However, contrary to the results of those studies, the band at 687 nm in the red region and the bands at 840, 841, 1004, 1083, 1271, 1222, and 1281 nm in NIR are also important for estimating the NDF in our study. Furthermore, our study shows that except for the bands in the NIR and red regions, the red-edge (704 nm) and SWIR (1449, 1485, 1522, 1744, 2091, 2107, 2246, 2276, and 2303 nm) regions are also important for predicting the NDF content. Although Zhao *et al.* [18] also successfully selected the bands in the SWIR region, they missed the red-edge region. In addition, the specific bands contained in the SWIR region (1635 and 1695 nm) selected by their study are also largely different from what we find (1449, 1485, 1522, 1744, 2091, 2107, 2246, 2276, and 2303 nm). Overall, the bands in the NIR and red regions are important for estimating the NDF content, and we find that the bands in the SWIR and red-edge regions also play a crucial part in estimating the NDF content.

Besides, not only does the sensitive spectral bands play an important part in predicting the NDF content, but also the VIs are the indispensable variables that can reflect the growth condition of vegetation and can thus be used as surrogate indicators for the growth and health condition of vegetation [75]–[77]. To reduce the number of bands, Safari *et al.* [42] re-combined the spectral data according to WorldView-2 satellite broadbands and calculated the NDSI. The R^2 of the inversion model for the ADF content based on the ultrasonic sward height and NDSI reached 0.62. These results indicated that VIs could improve the accuracy of forage nutrient content estimation, agreeing with ours. However, our study finds that in addition to NDSI, SR is an important VI for predicting the NDF and has a significant influence on the NDF.

2) Contribution of EFs to the Inversion of the Forage NDF Content: EFs such as topography, soil, meteorology, and vegetation indirectly or directly affect the spatial distribution of

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forage nutrients [78]–[81]. The topographical factors (profile curvature, elevation, aspect, plan curvature, and slope) can influence the hydrothermal conditions and their distribution, which will affect the surface vegetation structure and spatial distribution [82], [83]. Using the Kruger National Park in South Africa as a study area, Knox *et al.* [13] collected samples in dry and wet seasons and predicted the forage ADF using the MLR method based on ancillary data (i.e., slope, aspect, elevation, and species) and spectral data. The results of their analysis based on a leave one cross-validation (LOOCV) showed that the R^2 of the inversion model reached 0.36–0.49 and plant species and soil types contributed significantly to the ADF content. Using hyperspectral data (the interval of the bands transformed into 21 nm) as well as cumulative growing degree units and canopy height measured by Light Detection and Ranging as variables, Noland *et al.* [41] estimated the alfalfa NDF content using stepwise regression. These three types of variables achieved good results in predicting the NDF content, with an R^2 of 0.76, which indicated that the combination of spectral data, vegetation biophysical indicators, and temperature could improve the estimation of the forage NDF. Gibson *et al.* [84] researched the influence of different precipitation gradients on the forage nutrient contents (e.g., crude protein and crude fat) and showed that forage nutrient contents were different under different precipitation conditions, indicating that soil moisture had a significant effect on the forage nutrient content.

The EFs selected in our models include the plan curvature, SOL, precipitation, Clay1, and grass height, which are consistent with the conclusions described above. Fig. 5 shows that, among the five factors, the plan curvature contributes the most, with a proportion of 5.74%, followed by the grass height, SOL, Clay1, and precipitation, with contribution ratios of 4.83%. 4.39%, 2.37%, and 1.00%, respectively. In addition, as seen from the performance of the 87 models constructed in the study, the 12 models (R^2 of 0.37–0.55, and RMSE of 4.07%–4.85%) based on every type of the four spectral data combining with the EFs (OR-EFs, FD-EFs, CR-EFs, and Log-EFs) outperform the 12 models (R^2 of 0.35–0.51 and RMSE of 4.31%–4.92%) constructed by every type of four spectral data only (OR, FD, CR, and Log). Thus, although there are differences in specific EFs for the inversion of the fiber components in different ecoregions, it is certain that the addition of EFs plays a crucial part in improving the estimation accuracy of the fiber components.

D. Prospects

In this study, limited to the number of samples and the resources, the inversion models of the forage NDF content are constructed using all of the samples collected throughout the entire growth period over four years. Models are not constructed based on different growth stages (e.g., boot stage from May to June, peak growth from July to August, and dormancy from September to October). Therefore, in future studies, additional samples should be collected in the boot stage and the dormancy stage, and changes in the contents and digestibility of forage fiber components at different growth stages should be analyzed to construct inversion models for different growth stages, which could thereby offer an important way to improve the accuracy of the existing models.

Having followed the traditional space and satellite remote sensing, UAV remote sensing technology is regarded as the third generation of remote sensing technology and has increasingly become a research hotspot [85]–[87]. UAV can be used to quickly acquire near-surface information and complete the collection, processing, and analysis of remote sensing data, showing the advantages of flexibility, safety, and economy [88], [89]. The combination of UAV and multispectral remote sensing, which can improve the efficiency of traditional ground-based observations and expand the range, number, and spatiotemporal representativeness of samples, is the key to constructing high-accuracy models of nutrient contents of natural grassland with wide spatial distribution and complex species diversity [90], [91]. Therefore, investigation of the methods for monitoring the quality and nutritional status of alpine grassland via UAV and satellite technologies is very promising and one of the main directions for our future research.

V. Conclusion

The accuracy of the multifactor non-parametric inversion models of the forage NDF content in natural alpine grassland has been evaluated by combining the reflectance of original and three transformed spectra (CR, FD, and Log) with VIs and EFs. The results are summarized as follows.

1) By comparing 87 models based on the reflectance of feature bands of different data types combined with VIs and EFs, the RF models can invert the forage NDF more accurately than the ANN and SVM models. In addition, the FD-Log-VIs-EFs-RF model estimates the NDF content the most accurately, with an R^2 of 0.62 and an RMSE of 3.98%.

2) VIs constructed by the OR spectral bands, FD, and Log spectra are the main spectral variables used to estimate the NDF content. The sensitive bands used in the best model include a Log spectral channel at 2276 nm and 13 FD spectral channels, which are mainly in the red-edge (704 nm), NIR (840, 841, 1004, 1083, and 1271 nm), and SWIR (1449, 1485, 1522, 1744, 2091, 2107, 2245, and 2276 nm) regions. The three involved VIs (SR16, SR18, and NDSI01) include five OR spectral bands (687, 1222, 1281, 2276, and 2303 nm). A total of 18 bands are involved, of which 11 bands are similar to the known feature absorption bands of nutrient components such as fiber, N, starch, and protein, and the other seven bands consist of six NIR channels (840, 841, 1083, 1222, 1271, 1281 nm) and a SWIR channel (1485 nm).

3) All EFs significantly affect inversion models of NDF content. The EFs in the best model include the topography (plan curvature), vegetation biophysics (grass height), meteorology (SOL and precipitation), and soil (Clay1). These variables contribute 5.74%, 4.83%, 4.39%, 1.00%, and 2.37%, respectively, to the NDF content, reaching a total of 18.34%.

4) Compared with EFs, the hyperspectral sensitive variables play a more decisive role in estimating the NDF. Among them, the contributions of the 13 FD spectral variables, the Log

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spectral variable, and the three VIs are 48.77%, 8.18%, and 24.71%, respectively.

Overall, machine-learning algorithms based on spectral data and EFs (topography, meteorology soil, and vegetation biophysical indicators) are an important way to improve NDF estimation of alpine grassland in the Tibetan Plateau. Although the FD-Log-VIs-EFs-RF model can be used to estimate precisely the forage NDF content in our study, there are still some limits. A promising direction of research would be to start from different growth stages and combine UAV and satellite images to improve the representativeness and prediction accuracy of the samples and thereby evaluate the forage nutrient contents of large natural grassland.

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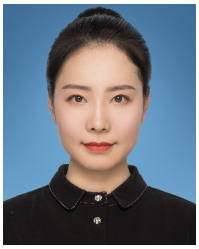
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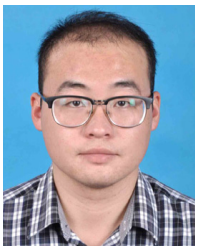
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